

Hypothesis Testing and Data Visualizations

What is statistics?

- Define research question, study design, data collection, summarize the data, analyze the data, generalize to the population, and communicate findings
- *Units/Subjects*: Entities on which data is collected
- *Variable*: Recorded characteristics for unit/subject
- *Population*: Group of interest for study
- *Sample*: Subset of population to study
- Results are generalized to the studied population

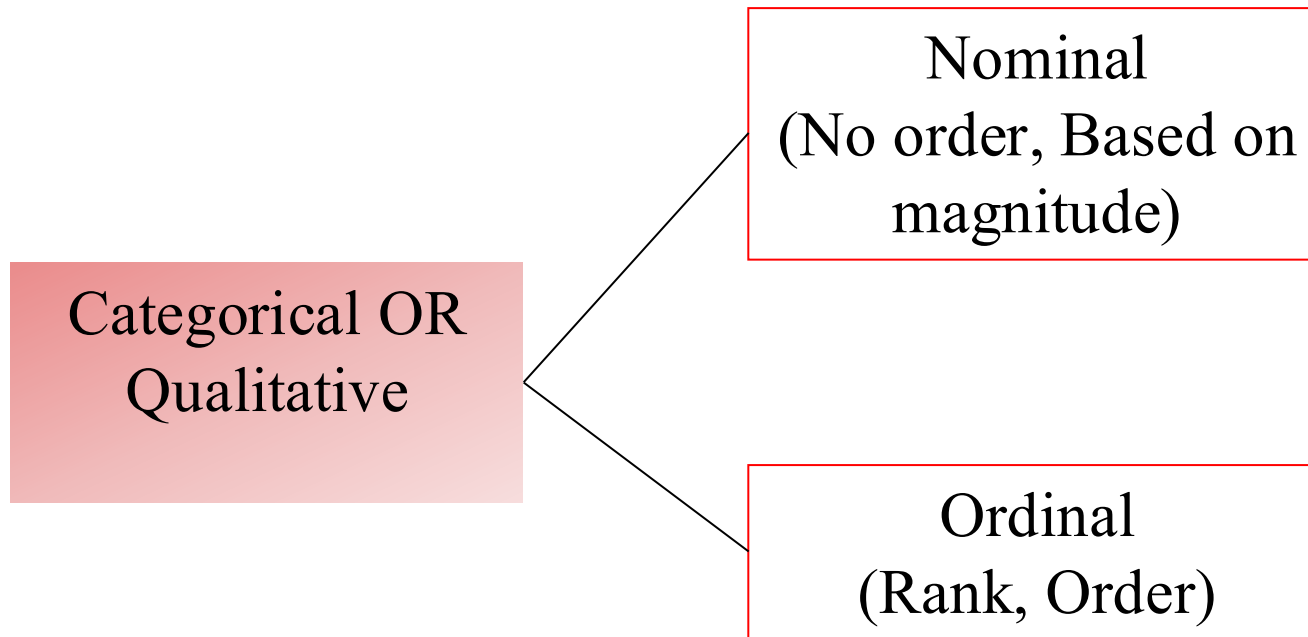
Example

- If university students exercise regularly, does that lower the risks of depression?
- Sample= 5000 university students from IITR
- Ask survey questions
- Variables: Exercise (y/n)
- Depression (y/n)
- (population stats): The true (population) difference in depression rates for those who exercise Vs those who don't.
- (sample stats): The difference in depression rates for those who exercise vs those who don't in the sample

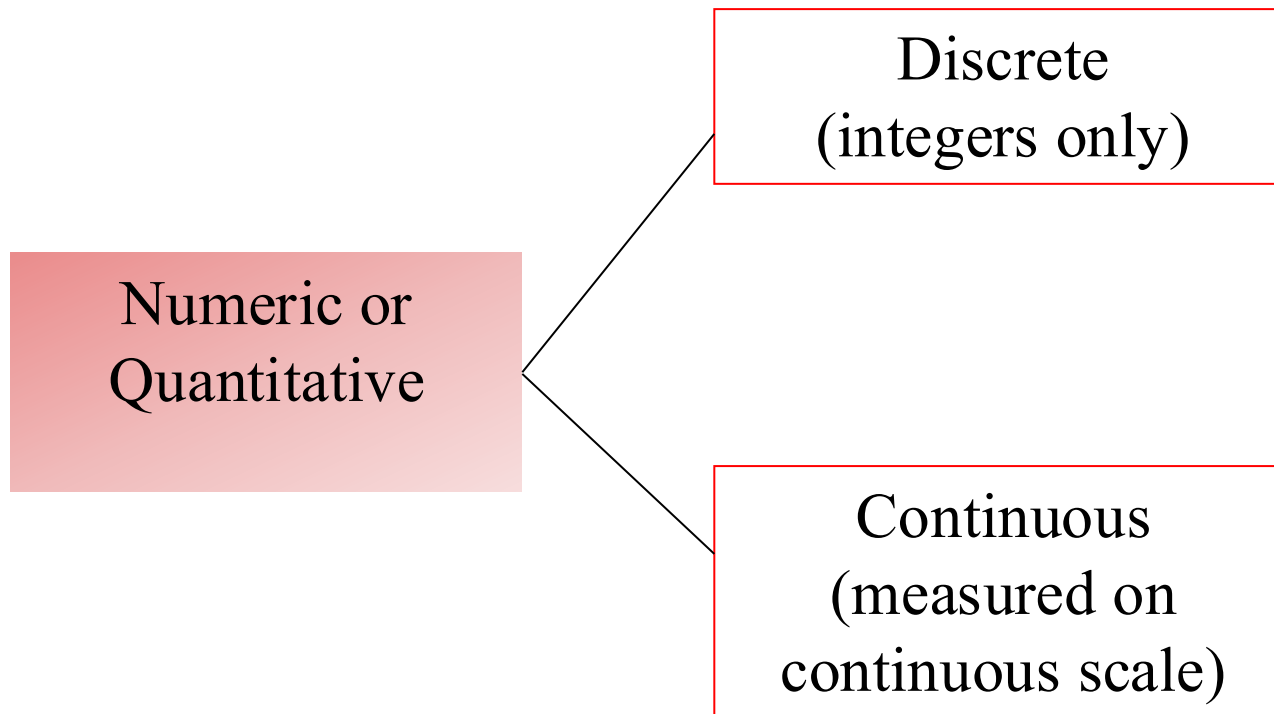
Continued...

- External validity: Is the result from our study generalizable to the general population of university students?
- Internal validity: Is the estimate within our study biased?

Variables and Types of Variable



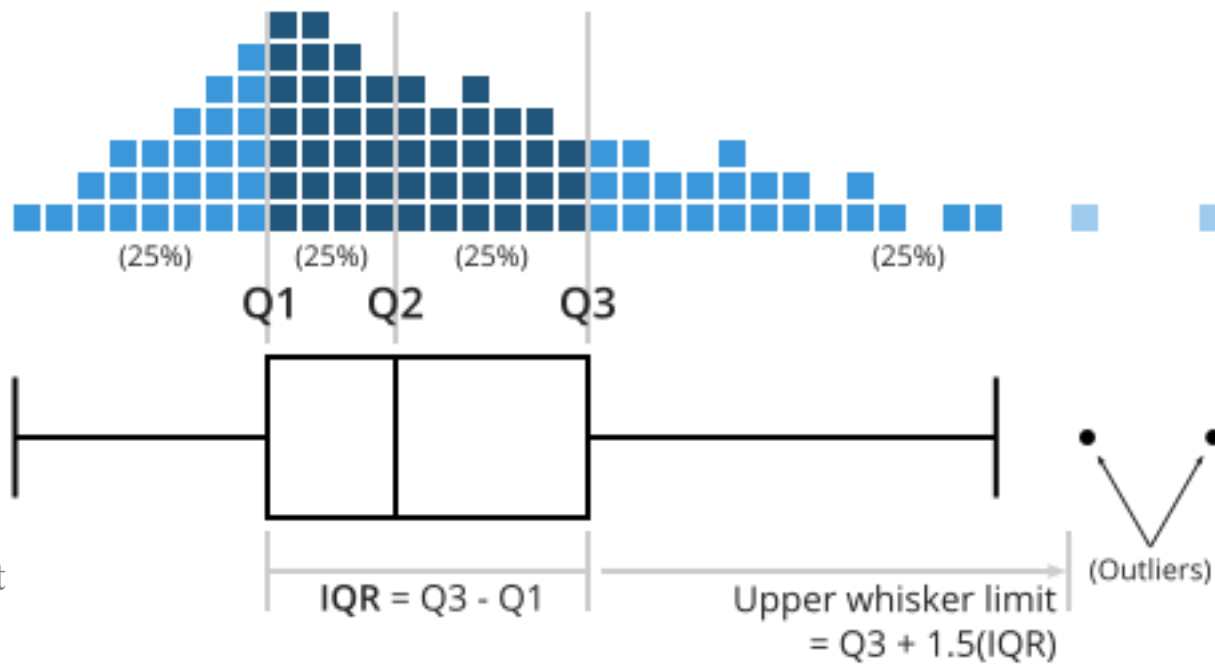
Variables and Types of Variable



Box Plot

- Used to visualize the distribution for numeric variables
- A box plot (aka box and whisker plot) uses boxes and lines to depict the distributions of one or more groups of numeric data.
- Box limits indicate the range of the central 50% of the data, with a central line marking the median value.
- Lines extend from each box to capture the range of the remaining data, with dots placed past the line edges to indicate outliers.

Box Plot



Q2 → Median

Continued...

Symmetric distribution



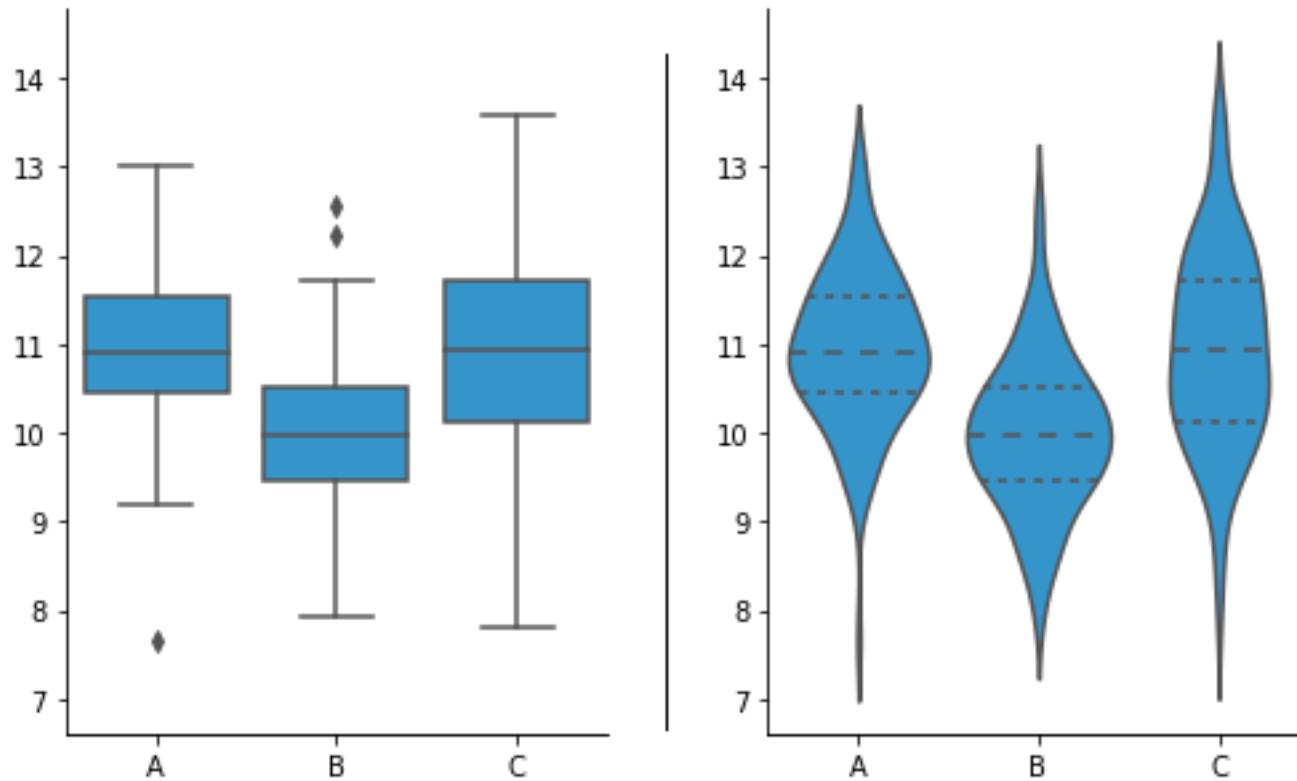
Skewed distribution



Violin Plot

- In a violin plot, each group's distribution is indicated by a density curve.
- In a density curve, each data point does not fall into a single bin like in a histogram, but instead contributes a small volume of area to the total distribution.
- Violin plots are a compact way of comparing distributions between groups.

Violin Plot

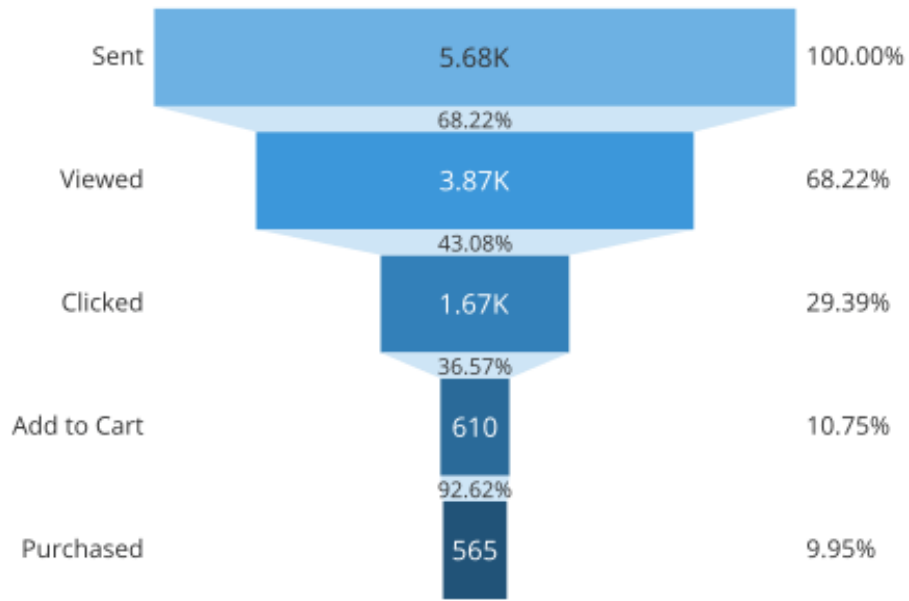


Funnel Chart

- A funnel chart is a specialized chart type that demonstrates the flow of users through a business or sales process.
- The chart takes its name from its shape, which starts from a broad head and ends in a narrow neck.
- The number of users at each stage of the process are indicated from the funnel's width as it narrows.

Funnel Chart

- The example funnel chart above depicts responses to a fictional email campaign regarding a special product offer.

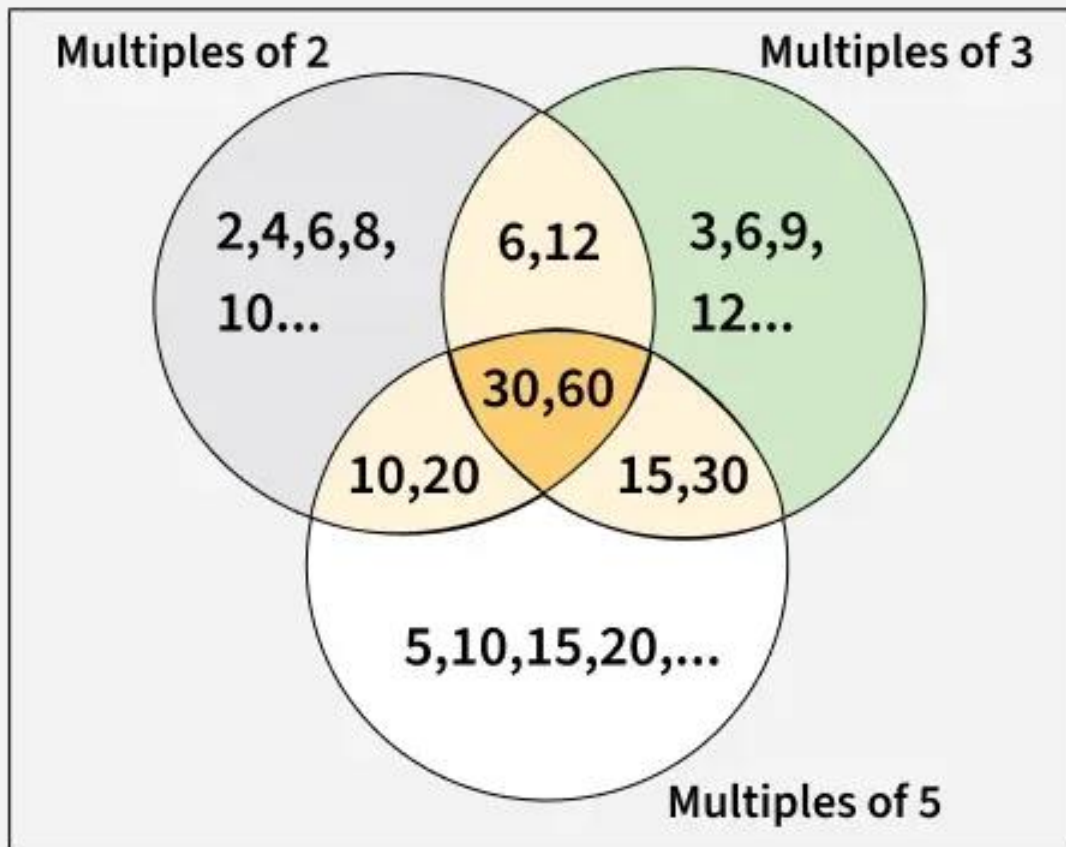


STAGE	USERS
Sent	5676
Viewed	3872
Clicked	1668
Add to Cart	610
Purchased	565

Venn Diagram

- Venn diagrams are visual tools used to illustrate the logical relationships between different sets of items.
- They use overlapping circles to show how sets intersect, share elements, or remain distinct.
- By categorizing items in this way, Venn diagrams make it easier to understand similarities and differences.
- Venn diagrams are also known as **set diagrams** or **logical diagrams**.

Example

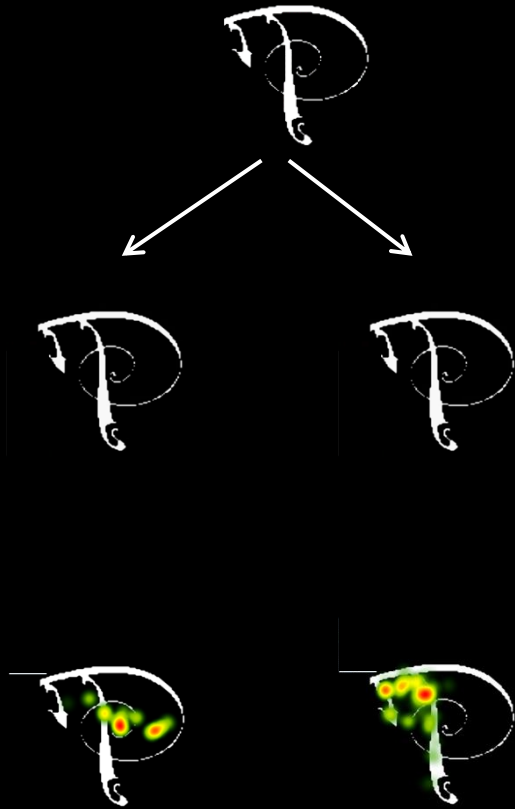


This demonstrates **relationships among three sets** and their **intersections**.

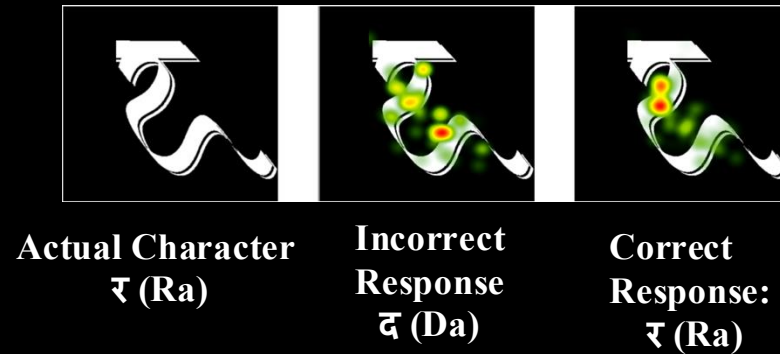
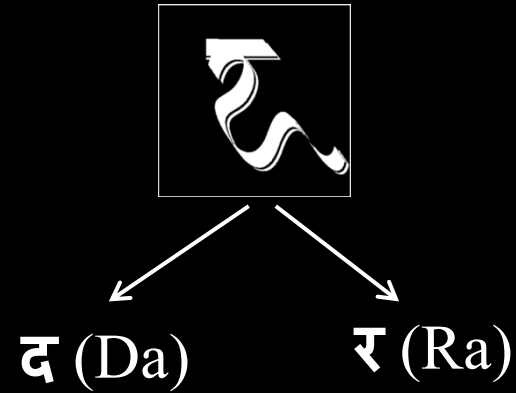
What do you see in the image?



Identify the character

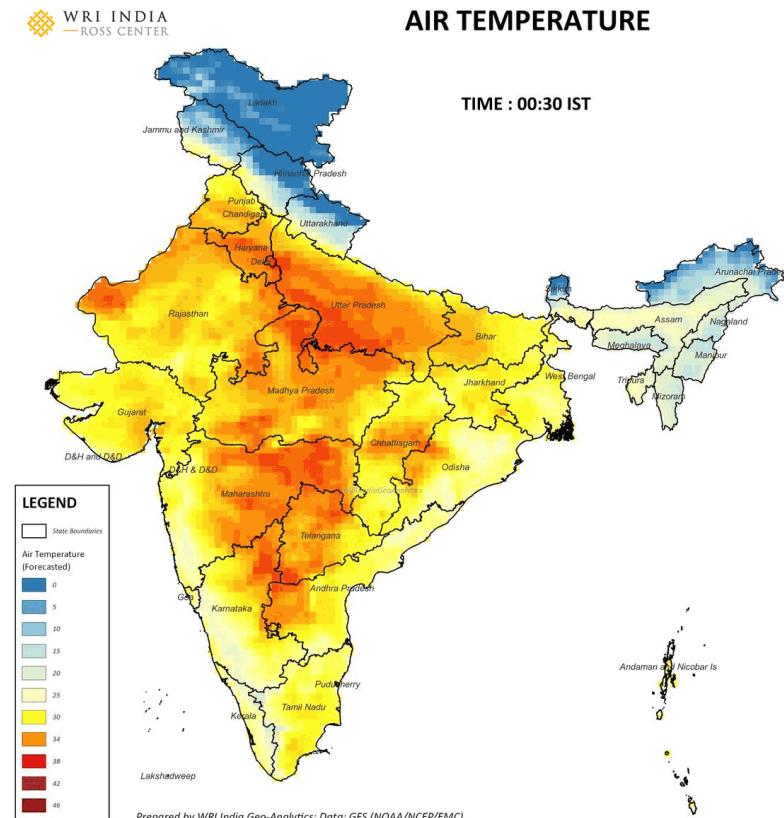


Not Recognized Recognized



Heat Maps

- Heatmap data visualization is a technique that uses color to represent data values.
- The most common color schemes range from warm colors (such as red) to cool colors (such as blue), with warm colors typically representing higher values and cool colors representing lower values.
- This visual representation allows for quick and intuitive understanding of complex data sets.

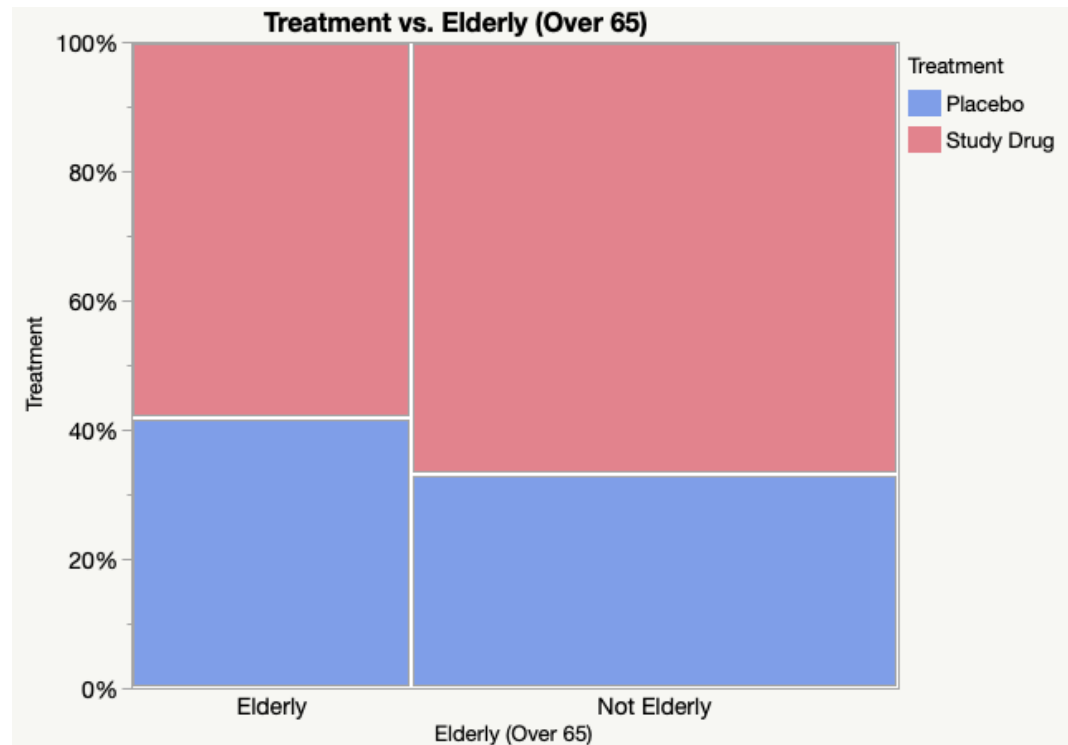


Mosaic Plot

- A mosaic plot is a special type of stacked [bar chart](#).
- For two variables, the width of the columns is proportional to the number of observations in each level of the variable plotted on the horizontal axis.
- The vertical length of the bars is proportional to the number of observations in the second variable within each level of the first variable.
- Mosaic plots help show relationships and give a visual way to compare groups.

Continued...

- Figure shows a mosaic plot for data from a clinical trial. The goal is to compare the distribution of patients over 65 years in both the placebo and study drug treatment groups.



The problem of hypotheses testing

- Statement of the Problem:
- Given a population (equivalently a distribution) with a parameter of interest, θ , (which could be the mean, variance, standard deviation, proportion, etc.), we would like to decide/choose between two complementary statements concerning θ . These statements are called statistical hypotheses.
- The choice or decision between these hypotheses is to be based on a sample data taken from the population of interest.
- The ideal goal is to be able to choose the hypothesis that is true in reality based on the sample data.

Situations where Hypotheses Testing is Relevant

- Example: A quality engineer would like to determine whether the production process he is charged of monitoring is still producing products whose mean response value is supposed to be μ_0 (process is in-control), or whether it is producing products whose mean response value is now different from the required value of μ_0 (process is out-of-control).
- Statement 1 (Null): $\mu = \mu_0$ (process in-control)
- Statement 2 (Alternative): $\mu \neq \mu_0$ (process out-of-control)

Some Situations ...

- **Example:** An engineer would like to decide which of two computer chip manufacturers (say, Intel and Motorola) is more reliable in producing computer chips. If we denote by p_1 the proportion of defective chips for Intel, and p_2 the proportion of defective chips for Motorola, then the goal is to decide between the following competing statements:
 - Statement 1 (Null): $p_1 \leq p_2$ (Intel is more reliable);
 - Statement 2 (Alternative): $p_1 > p_2$ (Motorola is more reliable).

Elements and Logic of Statistical Hypotheses Testing

- Consider a population or distribution whose mean is μ . To introduce the elements and discuss the logic of hypotheses testing, we consider the problem of deciding whether $\mu = \mu_0$, where μ_0 is a pre-specified value, or $\mu \neq \mu_0$.
- The **first step** in hypotheses testing, which should be done before you gather your sample data, is to set up your statistical hypotheses, which are the **null hypothesis** (H_0) and the **alternative hypothesis** (H_1).

The Statistical Hypotheses

- The null hypothesis, H_0 , is usually the hypothesis that corresponds to the **status quo**, the **standard**, the desired level/amount, or it represents the statement of “**no difference**.”
- The alternative hypothesis, H_1 , on the other hand, is the complement of H_0 , and is typically the **statement that the researcher would like to prove or verify**.
- These hypotheses are usually set-up in such a way that **deciding in favor of H_1 when in fact H_0 is the true** (called a **Type I error**) statement is a very serious mistake.

An Analogy to Remember

- Setting the null and alternative hypotheses has an analog in the justice system where the defendant is “presumed innocent” until “proven guilty.”
- In the court system, the null hypothesis corresponds to the defendant being innocent (this is the status quo, the standard, etc.).
- The alternative hypothesis, on the other hand, is that the defendant is guilty.
- Note that it is very difficult to reject the null (convict the defendant), and only “a proof (based on good evidence) beyond a reasonable doubt” will warrant rejection of H_0 .

The Hypotheses in our Problem

- For the problem we are considering, the appropriate hypotheses will be:
- $H_0: \mu = \mu_0$
- $H_1: \mu \neq \mu_0$.

Determine the Type of Sample Data that will be Gathered

- The second step is to determine what kind of sample data you will be gathering. Is it a simple random sample? A stratified sample?
- For the moment we will assume that a simple random sample of size n will be obtained, so the data will be representable by $X_1, X_2, \dots, X_n$, with $n \geq 30$.
- Also, determine if you know the population standard deviation σ . We assume for the moment that we do.

The Decision Rule

- The **decision rule** is the procedure that states when the null hypothesis, H_0 , will be rejected on the basis of the sample data.
- To specify the decision rule, one **specifies a test statistic**, which is a quantity that is computed from the sample data, and whose sampling distribution under H_0 is known or can be determined. Such a statistic measures the agreement of the sample data with the null hypothesis specification.
- For our problem, a reasonable choice for the test statistic is:

The Test Statistic

$$\bar{X} \quad \text{or equivalently, } Z_c = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}.$$

- The latter is a reasonable choice since it measures how far the sample mean is from the population mean under H_0 . The larger the value of $|Z_c|$ the more it will indicate that H_0 is not true.
- Furthermore, under H_0 , by virtue of the Central Limit Theorem, the sampling distribution of Z_c will be approximately standard normal.

When to Reject H_0 and its Consequences

- Having decided which test statistic to use, the next step is to specify the precise situation in which to reject H_0 . We have said that it is logical to reject H_0 if the absolute value of Z_c is large.
- But how “large” is “large”?
- For the moment, let us specify a **critical value**, denoted by C , such that if
- $|Z_c| > C$
- then H_0 will be rejected.
- Before deciding on the value of C , let us examine the consequences of our decision rule.

Possible Errors of Decision

- Remember at this stage that either H_0 is correct, or H_1 is correct. Thus, there is a “true state of reality,” but this state is not known to us (otherwise we wouldn’t be performing a test).
- On the other hand, our decision on whether to reject H_0 will only be based on partial information, which is the sample data.
- We may therefore represent in a table the possible combinations of “states of reality” and “decision based on the sample” as follows:

States of Reality and Decisions Made

		State of Reality	
		H_0 True	H_0 False
Decision Made Based on Sample Data According to Rule	Do not reject H_0	Correct Decision	Error in Decision (Type II error)
	Reject H_0	Error in Decision (Type I error)	Correct Decision

- In decision-making, there is therefore the possibility of committing an error, which could either be an error of Type I or an error of Type II.
- Which of these two types of error is more serious??

Assessing the Two Types of Errors

- From the table in the preceding slide, we have:
- **Type I error**: committed when H_0 is rejected when in reality it is true.
- **Type II error**: committed when H_0 is not rejected when in reality it is false.
- Just like in the court trial alluded to earlier, an error of Type I is considered to be a more serious type of error (“convicting an innocent man”).
- Therefore, we try to minimize the probability of committing the Type I error.

Setting the Probability of a Type I Error

- In trying to minimize, however, the probability of a Type I error, we encounter an **obstacle** in that the **probabilities of the Type I and Type II errors are inversely related**. Thus, if we try to make the probability of a Type I error very, very small, then it will make the probability of a Type II error quite large.
- As a compromise we therefore **specify a maximum tolerable Type I error probability**, called the **significance level**, and denoted by α , and choose the critical value C such that the probability of a Type I error is (at most) equal to α .
- This α is conventionally set to 0.10, 0.05, or 0.01.

Determining the Critical Value, C

- Let us now determine the critical value C in our test. Recall that our test will reject H_0 if $|Z_c| > C$.
- $P\{\text{Type I Error}\} = P\{\text{reject } H_0 \mid H_0 \text{ true}\} = P\{|Z_c| > C \mid H_0 \text{ true}\}.$
- But, under H_0 , Z_c is distributed as standard normal, so if we want $P\{\text{Type I error}\} = \alpha$, then we should choose the critical value C to be:
- $C = Z_{\alpha/2}$, which is the value such that $P\{Z > Z_{\alpha/2}\} = \alpha/2$.

The Resulting Decision Rule

- Given a significance level of α , for testing the null hypothesis $H_0: \mu = \mu_0$ versus the alternative hypothesis $H_1: \mu \neq \mu_0$, the appropriate test statistic, under the assumptions that (a) σ is known, and (b) $n \geq 30$ is given by:

$$\text{Reject } H_0 \text{ if } |Z_c| = \left| \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \right| > z_{\frac{\alpha}{2}}.$$

Data Gathering and Making the Decision

- Having specified the final decision rule, the next step is to gather the sample data and to compute the sample mean and the value of Z_c .
- If $|Z_c| > z_{\alpha/2}$ then H_0 is rejected; otherwise, we say that we “fail to reject H_0 .”
- **Note:** If σ is not known, then we could replace it in the formula of Z_c by the sample standard deviation S .
- The final step is to make the relevant conclusion.

On the Conclusion that One Could Make

- The final step in performing a statistical test of hypotheses is to make the conclusion relevant to the particular study, that is, not to simply say that “ H_0 is rejected” or “ H_0 is not rejected.”
- When H_0 is rejected, then either that a correct decision has been made, or an error of Type I has been committed. But since we have controlled the probability of committing a Type I error (set to α , which we could tolerate), then we can conclude in this case that H_0 is not true, and hence that H_1 is correct.

On Conclusions ... continued

- On the other hand, if we did not reject H_0 , then either we are making the correct decision, or we are making a Type II error.
- However, since we did not control for the Type II error probability (when we set the Type I error probability to be α , we “closed our eyes to the probability of a Type II error”), if we do not reject H_0 , we cannot conclude that H_0 is true. Rather, we could only say that we “failed to reject H_0 on the basis of the available data.”
- This is the basis of the saying that: “you can never prove a theory, you can only disprove it.”

Recapitulation: Steps in Hypotheses Testing

- **Step 1:** Formulate your null and alternative hypotheses.
- **Step 2:** Determine the type of sample you will be getting with regards to sample size, knowledge of the standard deviation, etc.
- **Step 3:** Specify your level of significance.
- **Step 4:** State precisely your decision rule.
- **Step 5:** Gather your sample data and compute the test statistic.
- **Step 6:** Decide and make final conclusions.

The p-Value Approach

- Another approach to making the decision in hypotheses testing is to compute the p-value associated with the observed value of the test statistic.
- By definition, the p-value is the probability of getting the observed value or more extreme values of the test statistic under H_0 .
- In our situation, the p-value would then be:
- $\text{p-value} = P \{ |Z| \geq |z_c| \}$ where z_c is the observed value of the test statistic.

Deciding Based on the p-Value

- If the p-value exceed 0.10, then H_0 is not rejected and we say that the result is not significant.
- If the p-value is between 0.10 and 0.05, we usually say that the result is almost significant or tending towards significance.
- If the p-value is between 0.05 and 0.01, we reject H_0 and conclude that the result is significant.
- If the p-value is less than 0.01 then H_0 is rejected and conclude that the result is highly significant.
- Or, we may compare the p-value with the level of significance: if it is smaller, reject H_0 .